

Fake News & Misinformation Detection System

End-to-end NLP Seminar Project

Seminar Project Team

HCMUS

June 18, 2026

Business Problem and Motivation

- Misinformation spreads faster than manual fact-checking can respond.
- Review teams need a fast first-pass triage system with evidence and explanations.
- The system is designed as a decision-support tool, not a fully autonomous censor.

Success metrics

Business: reduce manual triage time and reviewer workload. Technical: accuracy, F1, latency, uncertainty rate, and evidence quality.

Why NLP and What the System Does

- Input is unstructured article text, so semantic understanding is required.
- The system combines classification, retrieval, and decision calibration.
- Outputs: prediction, confidence, trust score, evidence, and explanation.

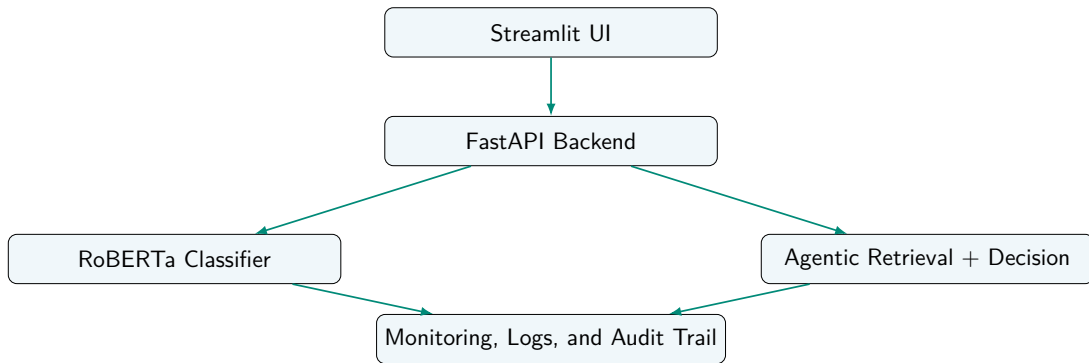
Model stack

TF-IDF + Logistic Regression
baseline

RoBERTa main classifier

Agentic retrieval and decision layer

System Architecture



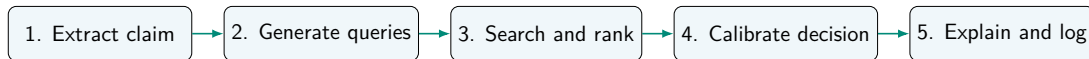
- English news-style text with binary labels: `fake` and `real`.
- Split strategy: 70% train, 15% validation, 15% test with stratification.
- Cleaning: HTML/URL removal, normalization, duplicate filtering, empty-row handling.
- Known risks: source bias, topic imbalance, and out-of-domain generalization gaps.

Model Selection and Evaluation

Model	Accuracy	Precision	Recall	F1
TF-IDF + Logistic Regression	0.9913	0.9938	0.9870	0.9904
RoBERTa	0.9991	0.9987	0.9997	0.9992

Trade-off

RoBERTa gives the best metrics, while the baseline remains easier to explain and cheaper to run.



Final states: REAL, FAKE, or UNCERTAIN when classifier and evidence disagree.

Recent Hardening and Corrections

- Claim-centric query generation now prioritizes the assertion over topic words.
- Domain-aware fallback was added for science and health claims.
- NASA direct search recovers authoritative `nasa.gov` evidence.
- Source credibility now scores subdomains like `en.m.wikipedia.org` correctly.
- Accepted and rejected evidence are logged for debugging and review.

- Demo claims now pass the retrieval smoke tests.
- Perseverance case now retrieves NASA evidence in live mode.
- Trump/Bannon case retrieves Reuters-like evidence and supports reviewable output.

What this means

The retrieval layer is much more reliable for seminar demos, but it still depends on search-provider quality.

Deployment Overview

- FastAPI backend exposes `/predict`, `/analyze`, `/agent`, `/health`, and `/version`.
- Streamlit frontend provides a web demo for direct interaction.
- Deployment targets: Render for backend and Streamlit Community Cloud for UI.
- Monitoring captures latency, confidence drift, uncertainty rate, and conflict rate.

- Privacy: mask common PII before inference and logging.
- Robustness: handle noisy, ambiguous, and out-of-domain claims explicitly.
- Ethics: false positives and false negatives can both harm users.
- Explainability: evidence, trust score, and rejected-evidence audit trail support human review.

Limitations and Future Work

- The classifier can still be overconfident on some true claims.
- Live retrieval is still sensitive to search noise and snippet quality.
- Stronger semantic entailment would improve evidence acceptance.
- Future work: better calibration, richer monitoring, and more domain-specific source routing.

Why stopping here is acceptable

The project already satisfies the seminar deliverables: end-to-end NLP, deployment, agentic behavior, monitoring, and responsible-AI discussion.

Key Takeaways

- Built a production-style fake news detection system with retrieval and explainability.
- Improved live evidence handling with claim-centric and domain-aware retrieval.
- Documented remaining limitations honestly for academic submission.

Q&A